**Industrial Internship Report on**

**" Quiz Game Project"**

**Domain:Python**

**Prepared by**

**[A.Revathi]**

| *Executive Summary* |
| --- |
| This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).<br><br>This internship was focused on a **Quiz Game** provided by UCT. We had to finish the project including the report in 6 weeks' time.<br><br>**My project title: Quiz Game Project.**<br><br>**Domain:Python.** |

**Problem Statement:**

**Description:** The quiz game is a Python project that quizzes users on various topics. It reads questions and answers from a file or database, presents them to the user, and keeps track of their score.

**Scope:** The scope of this project involves designing a user interface to display questions and collect user answers, implementing a database or file system to store quiz data, and developing a scoring algorithm to track the user progress and calculate their final score.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

**TABLE OF CONTENTS**

1 Preface ..... 3
2 Introduction ..... 8
2.1 About UniConverge Technologies Pvt Ltd ..... 8
2.2 About upskill Campus ..... 13
2.3 Objective ..... 16
2.4 Reference ..... 16
2.5 Glossary ..... 16
3 Problem Statement ..... 16
4 Existing and Proposed solution ..... 17
5 Proposed Design/ Model ..... 20
5.1 High Level Diagram (if applicable) ..... 21
5.2 Low Level Diagram (if applicable) ..... **Error! Bookmark not defined.**
5.3 Interfaces (if applicable) ..... **Error! Bookmark not defined.**
6 Performance Test ..... 22
6.1 Test Plan/ Test Cases ..... 22
6.2 Test Procedure ..... 22
6.3 Performance Outcome ..... 24
7 My learnings ..... 26
8 Future work scope ..... 27

# 1 Preface

## Summary of the whole 6 weeks' work.

➢ The primary focus of first week is to understand the one of the python projects from multiple project lists.

➢ The goal for the second week is to implement the second step of the quiz game project.

➢ This week, I completed half of the code for creating a graphical user interface using Tkinter.

➢ The goal for the third week is to implement the third step of the quiz game project.

➢ This week, I completed 65% of the code for creating a graphical user interface using Tkinter.

➢ I created def buttons(self), def display_options(self) and def display_question(self) functions.

➢ Fourth week, I completed the entire code for creating a graphical user interface using Tkinter.

**Need of relevant Internship in career development.**

**1.Practical Experience**:

This internship experience will provide you with valuable practical skills in game development, teamwork, problem-solving, and communication. It will also give you insight into the iterative nature of software development and the collaborative process involved in creating a game.

**2.Skill Development**:

An internship focused on a quiz game project provides a valuable opportunity to develop a diverse range of skills relevant to game development and software engineering, setting a strong foundation for future career growth in the industry.

**3.Industry Exposure**:

An internship focused on a quiz game project offers valuable exposure to the gaming industry, providing insights into game development processes, technologies, market dynamics, and professional practices. This exposure sets a solid foundation for pursuing a career in the gaming industry or related fields.

**4.Career Exploration**:

Participating in a quiz game project internship opens up various career exploration opportunities within the gaming industry and related fields, allowing you to discover your interests and strengths while gaining practical experience and valuable skills.

**5.Professional Development**:

Participating in a quiz game project internship offers a rich learning experience that contributes to your professional development, equipping you with valuable skills, experiences, and connections that will benefit your career in the gaming industry and beyond.

## Brief about project/problem statement.

**Project Titlle**: **Quiz Game**

**Project Overview:** The Interactive Quiz Game Application is a Python-based project designed to provide users with an engaging platform to test their knowledge across various topics. The application features a user-friendly interface, a diverse question database, and interactive gameplay mechanics aimed at enhancing user experience and learning.

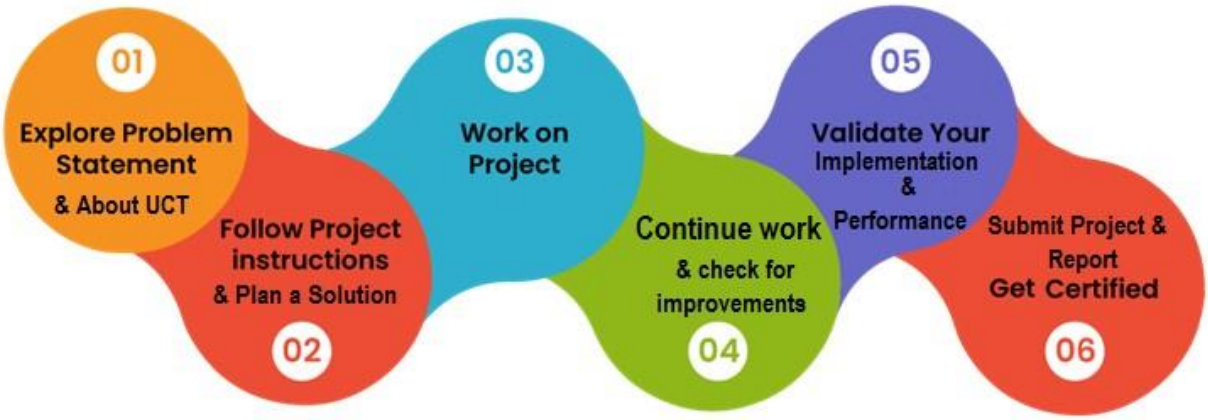
**Project Objective:**

1. Develop an interactive quiz game application using Python.
2. Create a user-friendly interface using Tkinter for seamless navigation.
3. Design a diverse question json file covering a range of topics.
4. Implement scoring and feedback mechanisms to track user performance.
5. Ensure data persistence for user profiles and quiz results.

**Opportunity given by USC/UCT**.

I would like to extend my sincere appreciation for providing me with the opportunity to intern at [ **UniConverge Technologies Pvt Ltd**]. My internship experience has been incredibly valuable, and I am grateful for the knowledge, skills, and insights I have gained during my time with your team.

**How Program was planned**




**My Learnings and overall experience.**

- Overall, the project serves as a valuable tool for users seeking to challenge their knowledge and engage in interactive learning experiences.
- Thank to all , who have helped you directly or indirectly.

**My message to my juniors and peers.**

UCT has offered an exceptional internship opportunity. We encourage you to enroll for the internship and take full advantage of this valuable opportunity.

# 2 Introduction

## 2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

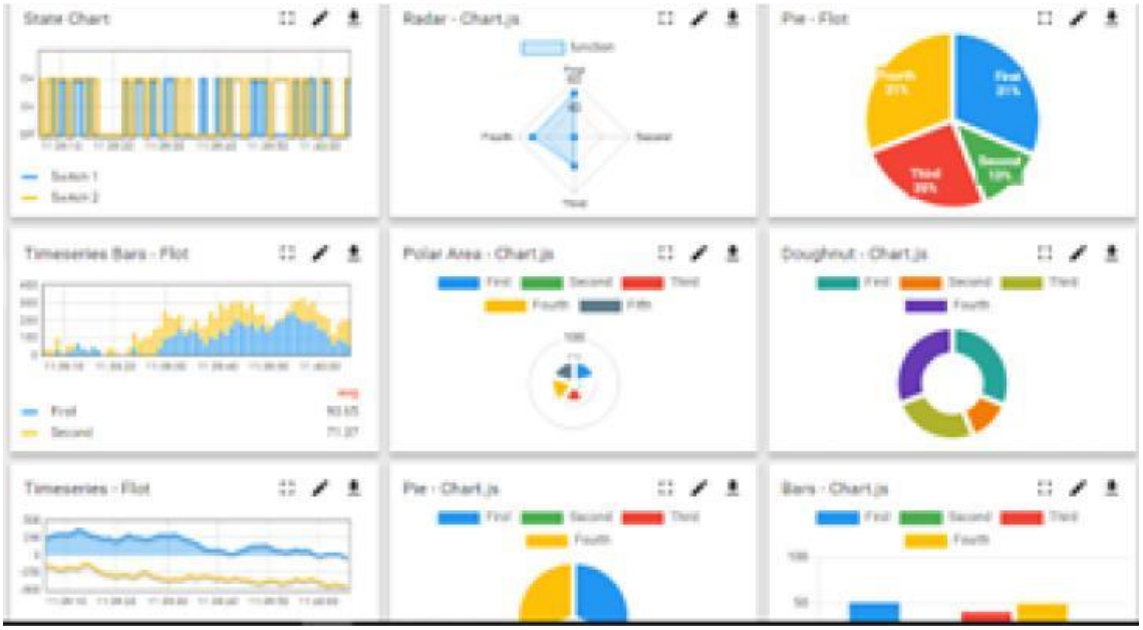
## i. UCT IoT Platform ( uct Insight )

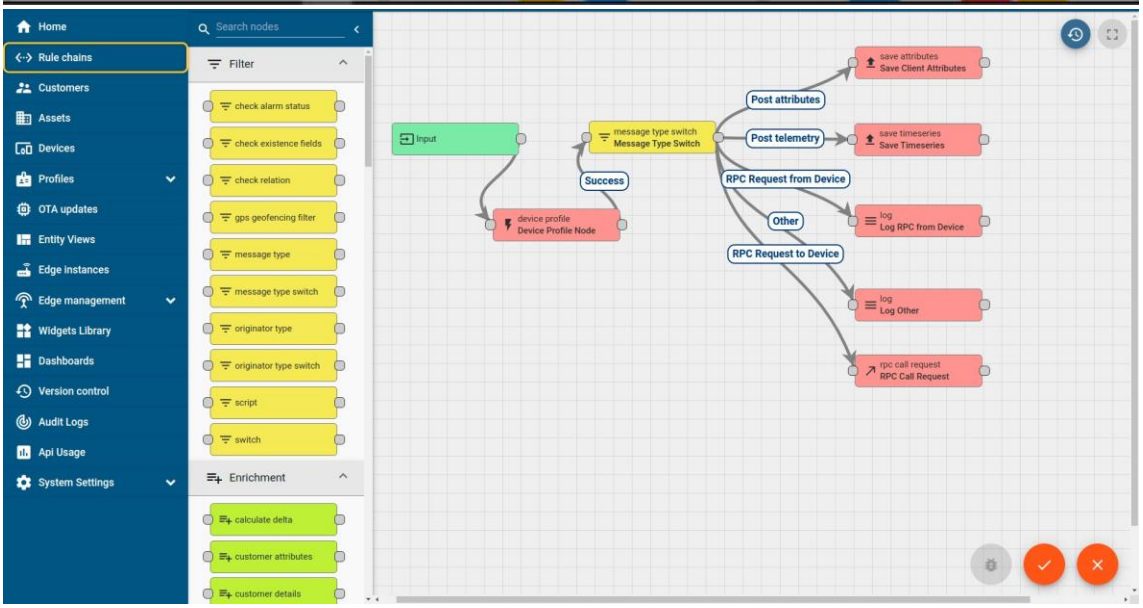
**UCT Insight** is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine


State Chart
Radar - Chart.js
Pie - Flot
Timeseries Bars - Flot
Polar Area - Chart.js
Doughnut - Chart.js
Timeseries - Flot
Pie - Chart.js
Bars - Chart.js


Home
Rule chains
Customers
Assets
Devices
Profiles
OTA updates
Entity Views
Edge instances
Edge management
Widgets Library
Dashboards
Version control
Audit Logs
Api Usage
System Settings
Search nodes
Filter
check alarm status
check existence fields
check relation
gps geofencing filter
message type
message type switch
originator type
originator type switch
script
switch
Enrichment
calculate delta
customer attributes
customer details
Input
message type switch
Message Type Switch
Success
device profile
Device Profile Node
Post attributes
Post telemetry
RPC Request from Device
Other
RPC Request to Device
save attributes
Save Client Attributes
save timeseries
Save Timeseries
log
Log RPC from Device
log
Log Other
rpc call request
RPC Call Request

## ii. Smart Factory Platform ( FACTORY WATCH )

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

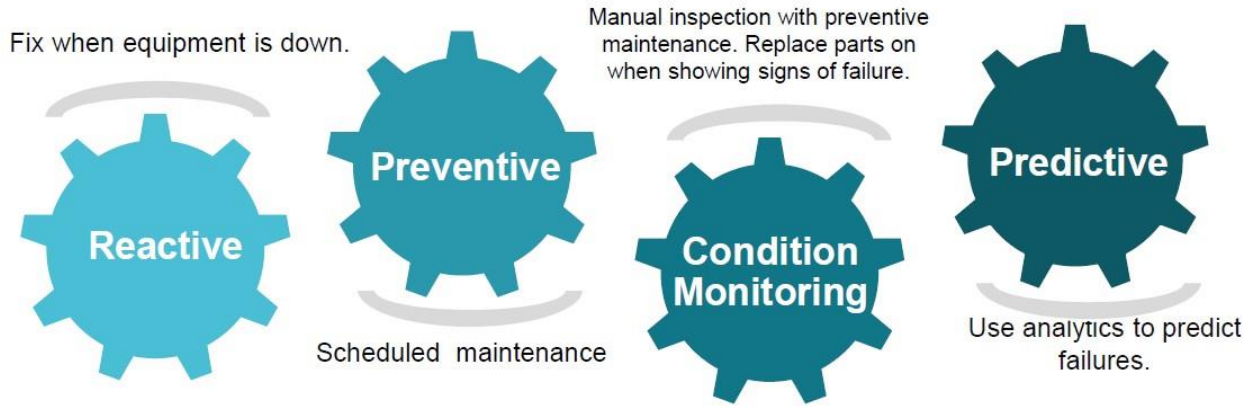
| Machine | Operator | Work Order ID | Job ID | Job Performance | Job Progress | | Output | | Rejection | Time (mins) | | | | Job Status | End Customer |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Start Time | End Time | Planned | Actual | | Setup | Pred | Downtime | Idle | | |
| CNC_S7_81 | Operator 1 | WO0405200001 | 4168 | 58% | 10:30 AM | | 55 | 41 | 0 | 80 | 215 | 0 | 45 | In Progress | i |
| CNC_S7_81 | Operator 1 | WO0405200001 | 4168 | 58% | 10:30 AM | | 55 | 41 | 0 | 80 | 215 | 0 | 45 | In Progress | i |

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc. **iv.**

Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.

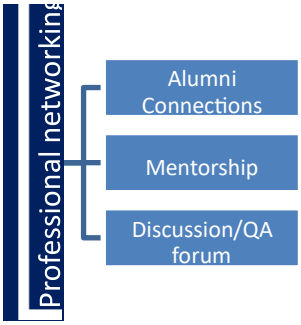


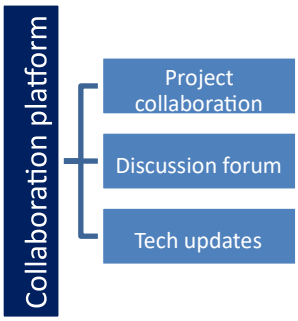

### 2.2 About upskill Campus (USC)

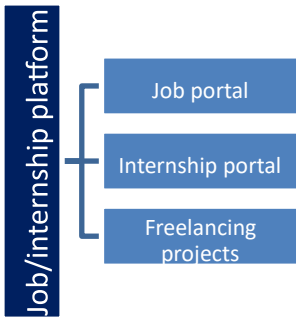
upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

Career growth/upskilling
Interview Preparation and skill building
upskilling Courses
Skill Assessment
Profile building


Professional networking
Alumni Connections
Mentorship
Discussion/QA forum


Collaboration platform
Project collaboration
Discussion forum
Tech updates


Job/internship platform
Job portal
Internship portal
Freelancing projects

## 2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

## 2.4 Objectives of this Internship program

The objective for this internship program was to ☛

get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

## 2.5 Reference


[1] Stack Overflow. https://stackoverflow.com/

[2] Official Tkinter Documentation-https://docs.python.org/3/library/tk.html

[3] GeeksforGeeks- https://www.geeksforgeeks.org/


## 2.6 Glossary

| Terms | Acronym |
|---|---|
| **GUI** | Graphical User Interface |
| **JSON** | JavaScript Object Notation |
| **Tkinter** | Toolkit Interface |
| **UI** | User Interface |
| **IDE** | Integrated Development Environment |

# 3 Problem Statement:

**Description:** The quiz game is a Python project that quizzes users on various topics. It reads questions and answers from a file or database, presents them to the user, and keeps track of their score.

**Scope:** The scope of this project involves designing a user interface to display questions and collect user answers, implementing a database or file system to store quiz data, and developing a scoring algorithm to track the user progress and calculate their final score.

## 4 Existing solution

**Mobile Quiz Apps:**

- **Description**: The mobile quiz app is a user-friendly application designed to provide an interactive and engaging platform for users to participate in quizzes on various topics.

- **Limitations:** The app's effectiveness relies on the availability of a diverse range of high-quality quiz content. If the content is limited or not regularly updated, users may lose interest over time.

**Custom-Built Quiz Applications**:

- **Description**: Custom-built quiz applications are bespoke software solutions designed to facilitate the creation, management, and delivery of quizzes tailored to specific needs and objectives. These applications offer a range of features and functionalities to create engaging quiz experiences for users.
.
- **Limitations**: Custom-built quiz applications offer organizations the flexibility, control, and customization options needed to create engaging and interactive quiz experiences tailored to their specific needs and objectives.

**Traditional Paper-Based Quizzes**:

- **Description**: Traditional paper-based quizzes are a classic form of assessment conducted using printed questionnaires or test papers. These quizzes are typically administered in classrooms, training sessions, workshops, or other educational settings.

- **Limitations**: Advancements in digital assessment tools and platforms offer opportunities to overcome many of the limitations associated with paper-based quizzes, providing more efficient, interactive, and data-driven assessment solutions.

**Proposed Solution:**

**Scalable and User-Friendly Interface**:

- **Description**: A scalable and user-friendly interface is a key component of software applications, websites, or digital platforms designed to accommodate growth, handle increased user traffic, and provide a seamless and intuitive user experience.
- **Value Addition**: The interface will prioritize ease of use and intuitive design principles, ensuring accessibility for users of all skill levels. Responsive design techniques will optimize the user experience across web browsers, mobile devices, and other platforms, allowing users to access quizzes anytime, anywhere.

In summary, the proposed solutions aim to deliver a comprehensive and usercentric quiz game platform that offers unparalleled customization, scalability, usability, analytics, community engagement, and innovation. By addressing key pain points and offering value-added features, the platform seeks to redefine the way users

create, share, and engage with quizzes, ultimately empowering learners to achieve their educational goals.

### 4.1 Codesubmission(Githublink)

https://github.com/ARevathi2004/upskillcampus/blob/master/QuizGame.py

https://github.com/ARevathi2004/upskillcampus/blob/master/data.json

### 4.2 Report submission (Github link) :

https://github.com/ARevathi2004/upskillcampus/blob/master/QuizGame_A.Revathi_USC_UCT.pdf

## 5 Proposed Design/ Model

1.Requirements Gathering

2.Conceptualization and Ideation

3. Architecture Design

4. User Interface (UI) Design

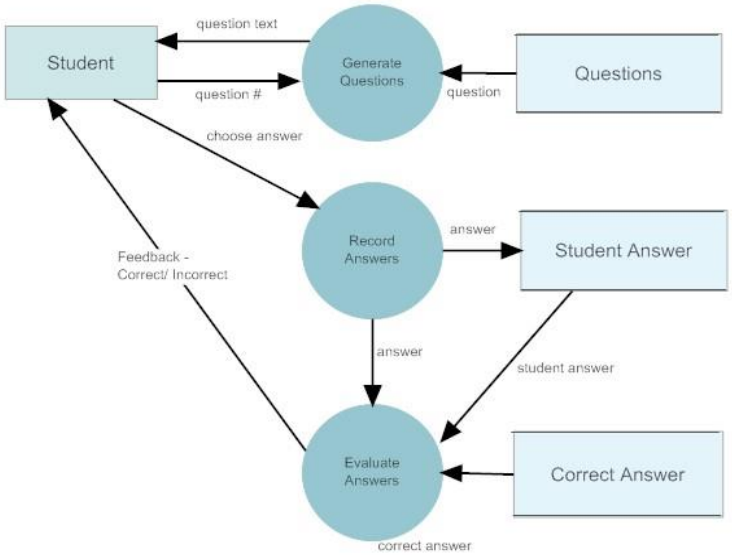
5. Development

6. Feature Implementation

7. Integration and Testing

### 5.1 High Level Diagram (if applicable)

Data Flow Diagram - Quiz Software



# 6 Performance Test

## 6.1 Test Plan/ Test Cases

1. **Test Strategy**

- Testing Approach (Automated)
- Testing Tools -Pycharm

2. **Test Scenarios**

- Quiz Creation
- Quiz Taking and Scoring
- User Interface and Navigation
- Error Handling and Validation
- Performance and Scalability

## 6.2 Test Procedure

1. **Test Case: Quiz Taking**

- Description: Test the functionality of taking a quiz and submitting answers.
- Steps:

1. Log in to the quiz game platform.

2. Select a quiz from the available list.
3. Answer each question within the specified time limit.
4. Submit the quiz and check for scoring and feedback.

2. **Test Case: Error Handling**

- Description: Verify that appropriate error messages are displayed for invalid inputs.
- Steps:

1. Enter invalid data (e.g., empty fields, incorrect format) in registration/login forms.
2. Submit the form and check for error messages.
3. Validate that error messages provide clear instructions for correction.

3. **Test Case: Performance Testing**

- Description: Conduct load testing to evaluate system performance under peak usage conditions.
- Steps:

1. Simulate multiple concurrent users accessing the quiz game platform.

2. Monitor system response time, throughput, and resource utilization.

3. Measure the scalability and stability of the system under increasing user loads.

### 6.3 Performance Outcome

1. **Response Time**: The response time for a quiz game project refers to the duration it takes for the game to react to user inputs or events, such as selecting an answer, loading a new question, or displaying feedback. By implementing these strategies and adopting a holistic approach to performance optimization, you can achieve low response times and deliver a responsive and engaging quiz game experience for users.
2. **Scalability**: Scalability is a critical aspect of any quiz game project, ensuring that the system can handle increased user traffic, data volume, and feature complexity as the user base grows.
3. **Throughput**: A quiz game project refers to the rate at which the system can process quiz-related operations, such as fetching questions, submitting answers, and updating game state, within a

given period. Maximizing throughput is crucial for ensuring that the quiz game can handle concurrent user interactions efficiently and provide a smooth and responsive user experience.

4. **Resource Utilization**: The efficient use of system resources, including CPU, memory, disk I/O, and network bandwidth. Monitoring resource utilization helps identify performance bottlenecks and optimize system configuration for optimal performance.
5. **Error Rates**: The frequency of errors, timeouts, and failures encountered by users during quiz sessions. Lower error rates indicate system stability and reliability, while higher error rates may indicate underlying issues that need to be addressed.
6. **Load Testing Results**: The results of load testing conducted to evaluate system performance under varying user loads and usage patterns. Load testing helps identify performance bottlenecks, scalability limitations, and areas for optimization.

## 7.My learnings

Technical Proficiency

1. Problem-Solving Skills
2. Communication and Collaboration
3. Adaptability and Resilience and Organization
4. Project Management
5. Continuous Learning and Growth Mindset
6. Professional Networking and Relationship Building
7.

## 8.Future work scope

1. **Additional Quiz Categories**
2. **Dynamic Question Generation**
3. **Customization Options**
4. **Integration with Social Media**
5. **Accessibility Features**
6. **Monetization Strategies**
7. **Integration with Social Media**
8. **Analytics and Insights**

Future work scope for a quiz game project can encompass various areas aimed at improving and expanding the game to enhance user engagement, functionality, and overall experience.

By focusing on these areas of future development, the quiz game project can evolve into a thriving and engaging platform that attracts and retains a loyal user base while remaining relevant in an ever-changing gaming landscape.